

FLOODSENSE

FloodSense: An Android Application Using an Expert System for Flood Susceptibility Assessment and Pre-Event Preparedness in Bacoar City

REVISED FEATURE SPECIFICATIONS & COMPLETE SYSTEM ARCHITECTURE

Development Planning and Implementation Guide

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Document Status

This is a revised Feature Specification, not a Software Requirements Specification (SRS). It defines the intended features, user/admin behavior, recommended implementation architecture, and deployment stack. It does not claim that the application, Expert System, database, web administration system, or external institutional datasets are already implemented or approved.

1. Document Overview

1.1 Purpose

This document provides the complete revised feature specification for FloodSense after expanding the system from a mobile-only flood awareness application into a deployable mobile-and-web platform. It keeps the original thesis purpose - flood susceptibility assessment and pre-event preparedness in Bacoar City - while defining the mobile user application, a new web-based administration system, a dynamic Geographic Information System (GIS) map, a rule-based Expert System, a Decision Support System (DSS), evacuation-center management, authentication, backend services, data management, and deployment architecture in implementation-ready detail.

The document is intentionally written as a feature and implementation guide rather than an SRS. Each feature explains what the user or administrator sees, what inputs are required, what processing occurs, how information flows to other modules, what data must be maintained, and what technology is recommended so the development team can translate the specification into code with minimal ambiguity.

1.2 Revised Product Scope

FloodSense will consist of two connected applications supported by one secured backend and one spatial database: (1) an Android application for residents, and (2) a web administration application for authorized maintainers. The Android application is the public-facing system. The web administration application is the maintenance and publishing environment for geographic data, Expert System rules, DSS content, evacuation centers, rainfall parameters, disclaimers, source metadata, and other controlled content.

The revised system no longer treats the susceptibility map as a fixed pre-rendered image. Geographic zones are stored as editable spatial polygons. Their fill colors are computed from the Expert System result for the active rainfall intensity and duration. When the user changes the rainfall scenario, the same zone geometries remain on the map while their susceptibility classifications and colors are refreshed.

1.3 Abbreviations and Technical Terms

To avoid confusion during thesis review and development, every abbreviation is written in full the first time it appears in the document and the abbreviation is placed in parentheses. After that first definition, the shortened form may be used. The following table provides the principal terms used throughout this specification.

Term / Abbreviation	Meaning in FloodSense
Application Programming Interface (API)	The secured server interface used by the Flutter application to request data and submit actions to Django.
Decision Support System (DSS)	The preparedness-support component that uses the Expert System result and the user's situational answers to provide approved guidance.
Geographic Information System (GIS)	The geographic data and map capability used to store, query, edit, and display zones, polygons, points, and evacuation centers.
Global Positioning System (GPS)	Satellite-based device positioning. FloodSense does not require GPS for Check My Area because the user manually places a pin.
Minimum Viable Product (MVP)	The smallest complete thesis implementation that proves the core research contribution and required features.
JavaScript Object Notation (JSON)	A structured data format used by the API.
GeoJSON	A JSON-based format for geographic points, lines, and polygons used when exchanging map geometry.
Django REST Framework (DRF)	The Django extension used to build the mobile API.
Object-Relational Mapper (ORM)	Django's model layer for reading and writing relational database

Term / Abbreviation	Meaning in FloodSense
	records using Python classes.
Hypertext Transfer Protocol Secure (HTTPS)	Encrypted network transport used by the production mobile application and admin website.
Android App Bundle (AAB)	Google Play's preferred Android release package generated from Flutter.
JSON Web Token (JWT)	A signed access-token format that may be used for expiring mobile sessions if the team selects it during implementation.
Bacoor Disaster Risk Reduction and Management Office (BDRRMO)	Primary intended institutional expert/data partner referenced by the existing FloodSense documents.
Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA)	Source of the public rainfall-classification reference used by the project, subject to final verification of thresholds and wording.
Representational State Transfer (REST)	The web-service style used by the mobile API endpoints.
User Interface (UI)	The visible controls, screens, forms, and map interactions used by residents or administrators.
Uniform Resource Locator (URL)	A web address, such as an API endpoint or external navigation link.
Comma-Separated Values (CSV)	A simple tabular text-file format that may be used for approved data import/export.
Spatial Reference System Identifier (SRID)	The numeric identifier describing the coordinate reference system used by spatial data.
European Petroleum Survey Group (EPSG) code	A commonly used identifier system for coordinate reference systems; EPSG:4326 identifies WGS 84 geographic coordinates.
World Geodetic System 1984 (WGS 84)	A global geographic coordinate reference system commonly used for latitude/longitude exchange.
Cross-Site Request Forgery (CSRF)	A browser-request attack class; Django provides CSRF protection for state-changing web administration forms.
Simple Mail Transfer Protocol (SMTP)	Standard protocol used by the production server to send password-reset/verification emails through a configured email provider.
Multi-Factor Authentication (MFA)	Optional additional administrator login factor if later required by the institution.
Frequently Asked Questions (FAQ)	A structured public help/content section when used.
Internet Protocol (IP)	Network-address information that may appear in security/audit logs when legally and operationally appropriate.
Geospatial Data Abstraction Library (GDAL)	Native geospatial library commonly used by GeoDjango for spatial data handling.
Geometry Engine - Open Source (GEOS)	Native geometry library used by GeoDjango for geometric operations.
PROJ	Coordinate-transformation library used in geospatial software; treated as a library name rather than an abbreviation that must be expanded.
Artificial Intelligence (AI)	A possible future conversational layer only; it is not required for the core FloodSense Expert System or DSS.

1.4 Primary System Actors

Actor	Platform	Primary Responsibility
Resident / Mobile User	Android application	Select rainfall scenarios and locations; view dynamic susceptibility results; use Check My Area; receive Expert System explanations; follow DSS preparedness guidance; locate verified evacuation centers; manage personal profile and preferences.
Authorized Administrator	Web application	Maintain and publish the system knowledge and spatial data; manage Expert System rules, DSS decision trees, rainfall parameters, geographic zones, evacuation centers, source/provenance records, public content, versions, and audit history.
Subject-Matter Expert / Reviewer	Administrative workflow, not necessarily a separate account in the Minimum Viable Product (MVP)	Review proposed geographic assessments, rules, preparedness actions, and data sources before they are marked approved for public use. The exact institutional approval process must be agreed with the thesis adviser and partner offices.
External Map/Navigation Provider	External service	Provides basemap tiles and/or external turn-by-turn navigation. FloodSense remains responsible for its own susceptibility data and must not imply that the external provider validates flood safety.

1.5 Core Design Principles

- Explainability first: every public susceptibility result should be traceable to the active rule, geographic scope, scenario, data version, and plain-language explanation.
- Admin-maintained knowledge: rules, DSS questions, preparedness actions, evacuation centers, map geometries, and public text should be updateable from the web administration system without modifying the Flutter source code.
- Spatial accuracy follows source accuracy: FloodSense must not claim street-level precision when the approved source data support only barangay or zone-level classification.
- Scenario-based, not real-time forecasting: rainfall intensity and duration are planning inputs. FloodSense is not a live sensor, official warning issuer, or independent evacuation authority.
- Fail safely: unsupported areas, conflicting rules, stale data, unavailable services, or missing inputs must produce clear limitations rather than fabricated certainty.
- Minimize personal data: location pins used for assessment should be processed only as necessary and should not be stored by default.
- No conversational Artificial Intelligence is required for the core thesis: susceptibility assessment, the dynamic map, Check My Area, and the deterministic DSS operate without an AI assistant. Any future conversational feature must be treated as a separate enhancement.

1.6 Major Changes from the Earlier Feature Specification

Earlier Direction	Revised Direction
Pre-rendered susceptibility map	Dynamic GIS map using editable polygons and scenario-driven color styling.
Mobile-focused maintenance	Separate web administration system for system maintenance

Earlier Direction	Revised Direction
	and publishing.
Firestore-centric data storage	Managed PostgreSQL with PostGIS as the primary spatial and relational database.
Pre-computed lookup emphasis	Deterministic rule-based Expert System using decision tables and forward chaining; pre-computation may be used only as a performance cache, not as the definition of the reasoning model.
Username/password login wording	Email + password becomes the single login credential pair. A separate username login is removed because it is not required by the current feature set.
DSS content embedded conceptually in app	Database-driven DSS tree managed through the admin web interface so questions, branches, and actions can be revised without app code changes.
Evacuation centers as a static list	Admin-maintained spatial point dataset shown as dynamic map markers and ranked by proximity.

2. Complete System Architecture

2.1 High-Level Architecture

The revised FloodSense architecture deliberately uses one mobile framework, one server framework, and one primary relational/spatial database. The Android application is built with Flutter and Dart. The web administration system and backend are built in Python using Django. Django REST Framework (DRF) exposes the Application Programming Interface (API) used by the Android application. GeoDjango gives Django spatial data types and queries, while PostgreSQL with PostGIS stores the authoritative relational and geographic data. This design avoids maintaining separate FastAPI, Next.js, Firebase, or Supabase application layers that are not required by the current feature set.

The Android application and the administrator website never connect directly to PostgreSQL. All reads and changes pass through Django. Mobile users receive only published public data and can never edit geographic zones, Expert System rules, Decision Support System content, or evacuation-center records. Authorized administrators use the secured Django web system to maintain those records. This separation is essential because the public map is interactive, but only the administrative map is editable.

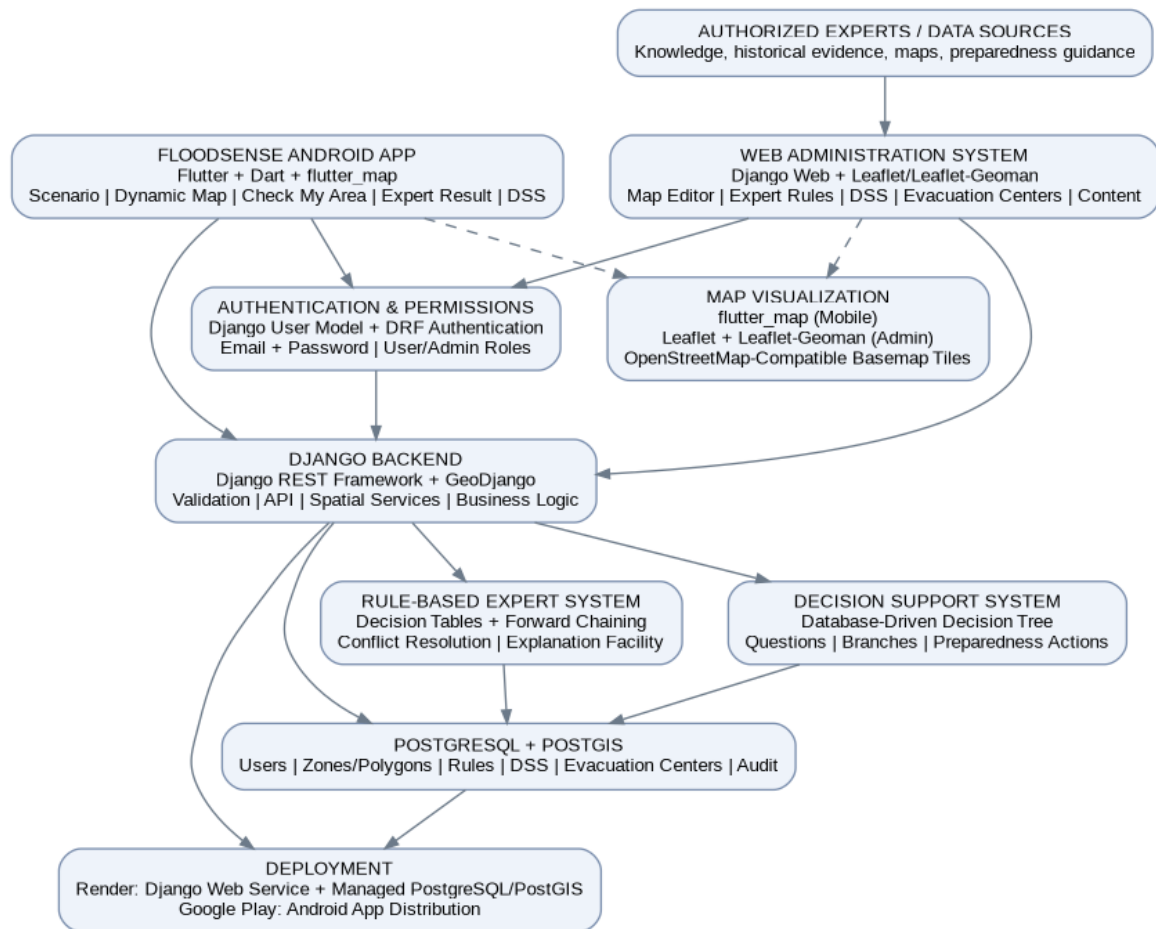


Figure 1. Recommended FloodSense production architecture.

2.2 Recommended Production Technology Stack

The following stack is the recommended baseline after removing technologies that duplicate responsibilities or increase the learning burden without providing a required FloodSense feature. Supporting packages may be added later when an implementation need is proven, but the core architecture should remain small and understandable.

Layer	Recommended Technology	Required Role in FloodSense
Android user application	Flutter + Dart	Builds the resident-facing Android application, including onboarding, authentication screens, dashboard, rainfall input, dynamic map, Check My Area, Expert System result screens, Decision Support System (DSS), evacuation resources, and account settings.
Mobile map library	flutter_map	Renders the interactive mobile map. Supports pan, zoom, polygon layers, markers, popups, layer visibility, dynamic polygon styling, and the user-selected location pin. The geographic data come from the FloodSense backend; flutter_map is the visualization layer.
Mobile API communication	Flutter http package	Sends Hypertext Transfer Protocol Secure (HTTPS) requests to the Django REST Framework API and receives JavaScript Object Notation (JSON) / GeoJSON responses. This is sufficient for the current API workload; a larger networking library is

Layer	Recommended Technology	Required Role in FloodSense
		not required unless later development proves a need.
Backend and web framework	Python Django	Provides the server-side application, built-in authentication, permissions, administrator website, data models, forms, validation, business logic, Expert System services, DSS services, content management, and deployment entry point.
Mobile API framework	Django REST Framework (DRF)	Creates secured REST-style API endpoints for the Flutter application. DRF handles request parsing, serialization, authentication integration, permissions, validation, and standardized API responses.
Spatial Django framework	GeoDjango	Extends Django models and queries with geographic fields and spatial operations. It connects the Django application to PostGIS and supports points, polygons, multipolygons, containment, distance, and other spatial queries needed by Check My Area and evacuation-center search.
Database	PostgreSQL	Stores normal relational data such as users, rainfall categories, rule metadata, DSS questions, preparedness actions, source records, versions, and audit history.
Spatial database extension	PostGIS	Extends PostgreSQL with geographic geometry/geography data types, spatial indexes, point-in-polygon operations, distance queries, and storage of barangay/susceptibility polygons and evacuation-center points.
Admin web interface	Django Admin + custom Django views/templates	Django Admin handles conventional record management. Custom Django pages are used where the generic admin is not enough, especially the dynamic map editor, rule-test workspace, DSS tree editor/preview, and publishing workflows. A separate Next.js/React frontend is not required.
Admin map library	Leaflet + Leaflet-Geoman	Displays the administrator map and provides interactive geometry tools for drawing, editing, dragging, moving vertices, deleting, and importing/exporting supported GeoJSON geometry. This is the main tool that lets authorized administrators manipulate FloodSense map objects visually.
Expert System engine	Custom Python rule engine inside Django	Implements a deterministic rule-based production system using decision tables and forward chaining. The knowledge comes from authorized experts and approved data sources; the engine evaluates that knowledge against the user scenario and returns an explainable susceptibility result.
Decision Support System engine	Database-driven decision tree inside Django	Uses the latest valid Expert System result plus the user's situational answers to select approved preparedness guidance. Questions, answers, branches, and actions are stored as versioned records so they can be maintained through the web administration system without changing Flutter code.
Authentication	Django custom user model + Django/DRF authentication	Uses email address + password as the actual user login credentials. The user model contains a server-side role/permission flag for resident versus administrator access. Django

Layer	Recommended Technology	Required Role in FloodSense
		sessions protect the admin web interface; DRF token-based authentication is used for the mobile API. JSON Web Token (JWT) authentication may be used through a small DRF extension if expiring access/refresh tokens are required for production.
Basemap tiles	OpenStreetMap-compatible tile service	Provides the visual street/place background underneath FloodSense data layers. FloodSense owns the susceptibility polygons, zone metadata, and evacuation-center overlays. For a public production release, the team must use a tile service whose usage policy supports the expected traffic rather than assuming unlimited use of a community tile endpoint.
External navigation	Android map deep link / url_launcher	Opens a verified evacuation-center destination in the user's installed navigation application. FloodSense supplies the destination coordinates but does not claim that the route is currently passable or flood-safe.
Source control	Git + GitHub	Stores the Flutter and Django source code, supports team collaboration, pull requests, issue tracking, release tags, and history. One repository with /mobile and /server directories is sufficient for the thesis team.
Django production server	Gunicorn + WhiteNoise on Render	Gunicorn runs the deployed Django application. WhiteNoise serves collected static assets for the admin website. Render documents this deployment pattern for Django and can connect the web service to managed PostgreSQL.
Server and web deployment	Render Web Service	Hosts the Django backend, DRF API, and web administration system together as one service, reducing infrastructure and deployment complexity.
Production database deployment	Render PostgreSQL with PostGIS enabled	Provides managed PostgreSQL and supports the PostGIS extension for geographic data. Keeping the Django service and database on one platform simplifies configuration for a student team.
Android deployment	Google Play Console	Flutter builds a signed Android App Bundle (AAB) for internal/closed testing and eventual production release. APK builds may be used for direct device testing, but the Play Store prefers the AAB format.

Primary Stack Decision

Core stack: Flutter + Dart + flutter_map for the Android application; Python Django + Django REST Framework (DRF) + GeoDjango for the server, API, and web administration system; PostgreSQL + PostGIS for relational and spatial data; Leaflet + Leaflet-Geoman for the editable administrator map; Render for Django and PostgreSQL/PostGIS deployment; Google Play for Android distribution. Next.js, FastAPI, Firebase, Firestore, Supabase, a separate object-storage service, Redis, Celery, Kubernetes, and a dedicated Expert System framework are not required by the current approved feature set.

2.3 Core Data Flow

Stage	Flow
Authentication	Mobile user/admin submits email + password -> Django authentication verifies credentials -> client receives session token -> token is attached to API calls -> Django backend validates token and role.
Assessment	User selects rainfall intensity + duration + location -> mobile calls assessment API -> PostGIS resolves the location to an approved zone -> Expert System loads active facts/rules -> forward chaining evaluates matches -> backend returns classification + explanation + version metadata.
Dynamic map	Mobile requests classifications for all visible/approved zones under the active rainfall scenario -> API returns zone IDs and results -> mobile applies color styles to zone polygons -> changing scenario repeats evaluation and restyles the map.
DSS	Expert result + scenario + location context starts the active DSS version -> user answers one decision node at a time -> backend or local approved tree follows the configured branch -> system returns preparedness actions and official-source reminders.
Evacuation centers	User location pin -> API/PostGIS retrieves verified active centers and computes proximity -> mobile shows markers and ranked list -> Get Directions opens external navigation to the selected verified center.
Admin publishing	Admin edits draft data/rules/tree/map -> validates and reviews -> publishes an approved version -> backend serves only the published active version to mobile users.

3. Complete Android User Flow

The user flow is designed so a resident can move from understanding the app, to selecting a rainfall scenario, to receiving an explainable susceptibility assessment, and finally to preparedness guidance and verified evacuation resources. The system does not require the user to understand GIS, millimeters per hour, rule engines, or database terminology.

Step	Feature	What Happens
1	Launch	User opens FloodSense. The app checks whether a valid local session exists and whether the active public configuration can be reached.
2	Onboarding and Disclaimer	First-time user reads the purpose, limitations, susceptibility meaning, geographic resolution, data date, and official-warning reminder, then confirms understanding.
3	Login / Registration	Existing user logs in with email address and password. New user registers with full/display name, email address, password, confirm password, home barangay, and required privacy/disclaimer acknowledgements.
4	Main Dashboard	User sees the active rainfall scenario, Expert System assessment entry point, dynamic map, Check My Area, Decision Support System (DSS), preparedness help,

Step	Feature	What Happens
		evacuation resources, data/methodology, and account settings.
5	Select Rainfall Scenario	User chooses an approved rainfall intensity category and duration. Plain-language descriptions are shown; technical rainfall values are supporting information.
6	Select or Check Location	User selects an approved zone/barangay from the map/list or positions a movable pin. The server resolves the selected point to the most specific supported geographic unit.
7	Expert System Assessment	Django builds the working facts and the Rule-Based Expert System evaluates only active approved rules using forward chaining. The result is Low, Moderate, High, Very High, Uncertain, or Insufficient Data.
8	Dynamic Susceptibility Map	Published zones are rendered as movable/zoomable map polygons and colored according to the active scenario. The user can tap zones and show verified evacuation-center markers.
9	Result Explanation	User sees the area, scenario, class, meaning, matched-rule explanation, source/version metadata, limitations, and official-source reminder.
10	Preparedness DSS	DSS uses the current Expert System result and the user's answers to follow approved preparedness branches and return appropriate actions.
11	DSS Preparedness Help	User selects a supported preparedness question/topic or searches approved keywords to receive validated guidance without requiring a conversational Artificial Intelligence system.
12	Evacuation Resources	User sees verified evacuation centers as map markers and a distance-ranked list. Get Directions opens an external navigation application.
13	Account and Preferences	User manages profile information, home barangay, optional default rainfall scenario, password actions, privacy information, and logout.

4. Detailed Android Feature Specifications

Feature 1 - Onboarding, Scope and Safety Disclaimer

The onboarding feature explains FloodSense before the user performs an assessment. Its purpose is not only instructional; it is also a safety control that prevents the user from interpreting a scenario-based susceptibility result as a live flood prediction, present road-safety statement, or independent evacuation order.

Aspect	Detailed Specification
Status	Core feature.
Trigger	Shown on first launch before the first assessment. It remains accessible from the dashboard under Data, Methodology and Disclaimer. A materially changed disclaimer version should require acknowledgement again.
Required Content	1) Purpose of FloodSense. 2) Meaning of flood susceptibility. 3) Explanation that the user selects a rainfall scenario for planning. 4) Supported geographic resolution. 5) Latest published data/rule version and update date. 6) Explanation of the Rule-Based Expert System. 7) Explanation of the Decision Support System (DSS). 8) Explanation that Check My Area uses a manually positioned pin and does not require Global Positioning System (GPS) permission. 9) Reminder to follow official BDRRMO/PAGASA advisories. 10) Privacy note describing what location information is processed.
Primary Action	I Understand / Continue. The action stores only the accepted disclaimer version and date, not any flood assessment.
Admin Control	Authorized administrators maintain the public onboarding text, disclaimer, official contact information, effective date, and published content version through the web administration system.
Failure Behavior	If the latest content cannot be loaded, the application may show the latest approved cached disclaimer with a visible date. If no approved version is available, assessment functions should remain unavailable rather than showing an unreviewed message.

Feature 2 - User Registration, Login and Session Management

FloodSense uses one unambiguous mobile login identifier: the user's email address. A separate username is not required for authentication because FloodSense has no social or public-identity feature that needs a unique username. Removing username login avoids duplicate-identifier logic and makes password recovery straightforward. The profile may store a full/display name, but it is not a credential.

Aspect	Detailed Specification
Actual Login Credentials	Email address + password.
Registration Fields	1) Full/display name - required for profile display only. 2) Email address - required, unique, and used as the login identifier. 3) Password - required. 4) Confirm Password - required and must match. 5) Home barangay - required convenience preference selected from the approved Bacoar list. 6) Accept Privacy Notice - required. 7) Accept FloodSense Disclaimer/Terms - required.
Email Rules	Trim leading/trailing spaces, compare using normalized lowercase form, validate basic email structure, and enforce uniqueness in the custom Django user model. The visible email may preserve the user-entered case, but authentication comparison should be normalized.
Password Rules	Use Django password validators. Recommended minimum: at least 8 characters and not a commonly used or entirely numeric password. Never store, return, display, or log plaintext passwords. Django stores a salted password hash.

Aspect	Detailed Specification
Confirm Password	Client checks match before submission; server repeats validation and never trusts client-only checks.
Home Barangay	Not an authentication credential and not a current-location claim. It is used only for convenience such as default map focus or profile context.
Registration Flow	User completes form -> Flutter validates required fields -> sends registration request over Hypertext Transfer Protocol Secure (HTTPS) -> Django validates again -> creates inactive/active account according to the chosen email-verification policy -> creates user profile -> returns success -> user logs in or continues if automatic login is approved.
Login Flow	User enters email + password -> Flutter sends credentials to the Django REST Framework (DRF) authentication endpoint -> Django verifies the password hash and account status -> server returns authenticated session/token information -> Flutter stores only the required token securely -> dashboard opens.
Mobile Session	Recommended production approach: expiring access token plus refresh token using a standard DRF JSON Web Token (JWT) package. A simpler DRF token may be used during early development, but the production design should support expiration/revocation. Tokens are never stored in source code.
Admin Session	The web administrator uses Django's server-side session authentication through the secured admin website. Admin access requires a staff/administrator role stored on the server.
Forgot Password	User enters registered email -> Django sends a time-limited password-reset link through configured email/Simple Mail Transfer Protocol (SMTP) service -> user creates a new password -> old reset token becomes invalid. The UI must not reveal whether arbitrary email addresses exist beyond what is necessary for the user experience.
Email Verification	Recommended before public production deployment. Registration sends a verification link; unverified accounts cannot use protected account features until confirmed. The team may postpone verification for a closed thesis pilot if approved, but the architecture should support it.
Logout	Revoke/expire the mobile session as supported, delete local authentication tokens, clear user-specific temporary assessment/DSS state, and return to Login.
Account Disabled	If an administrator disables an account, future protected API requests are rejected and the user is returned to login with a neutral account-unavailable message.
Error Messages	Examples: Invalid email format; Passwords do not match; Email already registered; Incorrect email or password; Account not verified; Account disabled; Network unavailable. Never expose database errors, stack traces, or whether a password was partly correct.
Security Boundary	No mobile user can obtain admin permissions by changing a local value. Administrator authorization is checked by Django on every protected web/API action.

Feature 3 - Main Dashboard

The dashboard is the resident's control center. It should explain the relationship between the rainfall scenario, Expert System, dynamic map, Check My Area, Decision Support System, and evacuation resources without requiring the user to understand technical implementation details.

Aspect	Detailed Specification
Dashboard Header	FloodSense logo/name; connection or data-state indicator; profile/account button; optional last-synchronization date.
Active Rainfall Scenario Card	Displays the selected rainfall intensity and duration. If no scenario is selected, the card becomes the primary action: Select Rainfall Scenario. If the scenario changes, any previous result is marked stale until re-evaluated.
Expert System - Assess Flood Susceptibility	Primary assessment feature. The card explains: FloodSense uses a Rule-Based Expert System whose knowledge base is built from authorized expert-provided/validated data and rules. It combines the selected rainfall scenario with the selected geographic zone and approved zone facts, evaluates applicable rules, and returns an explainable susceptibility classification.
What the Expert System Can Do	1) Evaluate an approved area under the selected rainfall intensity and duration. 2) Resolve the applicable geographic rule scope. 3) Apply expert-approved decision rules through forward chaining. 4) Produce Low, Moderate, High, or Very High susceptibility. 5) Return Uncertain or Insufficient Data instead of forcing a class when evidence/rules are inadequate. 6) Explain which rule and facts produced the result. 7) Generate consistent classifications for the map, results summary, and Check My Area. 8) Pass the validated result to the DSS. 9) Never learn from user behavior or override expert-approved rules automatically.
Dynamic Susceptibility Map	Opens the movable map. The user can pan, zoom, tap areas, view color-coded polygons, display evacuation-center markers, open the legend, and enter Check My Area pin mode. Colors are produced from the Expert System result for the current rainfall scenario.
Check My Area	Lets the user position a pin manually on the map. The backend identifies which approved polygon contains the point and evaluates that zone using the same Expert System.
Decision Support System (DSS)	Uses the latest valid Expert System result plus user-provided situational answers to provide approved pre-event preparedness support. It can also present supported help topics such as go-bag preparation, family communication, document protection, vulnerable household members, understanding the susceptibility result, and evacuation resources.
What the DSS Can Do	1) Start from the current susceptibility result and location context. 2) Ask structured situational questions. 3) Follow approved decision-tree branches. 4) Prioritize preparedness actions according to the user's answers. 5) Explain why an action is recommended. 6) Provide approved answers to supported preparedness topics. 7) Direct the user to verified evacuation resources when appropriate. 8) Remind the user to follow official authorities. 9) Never change the Expert System susceptibility class or independently issue an evacuation order.
Results Summary	Shows all supported areas and their classifications for the

Aspect	Detailed Specification
	active scenario. The same rule/map version must be used as the map.
Evacuation Resources	Shows verified active evacuation centers in map and list form. If a user pin exists, centers may be ranked by distance.
Data, Methodology and Disclaimer	Shows data-source names, geographic resolution, update dates, Rule-Based Expert System method, susceptibility meanings, DSS purpose, map limitations, privacy information, and official-source reminder.
Account and Preferences	Full/display name, email, home barangay, default rainfall scenario if desired, password reset/change, privacy information, and logout.
State Consistency	A result belongs to a specific geographic zone, rainfall scenario, map version, and rule-set version. If any of those change, the old result cannot silently remain current.

Feature 4 - PAGASA-Aligned Rainfall Scenario Input

Aspect	Detailed Specification
Purpose	Collect a standardized planning scenario without requiring a resident to estimate technical rainfall values manually.
Intensity Options	Baseline labels from the existing project documents: Light, Moderate, Heavy, Intense, and Torrential. Exact thresholds and wording must be verified against the final approved Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) reference before public use.
Duration Options	Baseline: 1 hour, 3 hours, 6 hours, 12 hours, and 24 hours. The final list must match the Expert System rule design and available data.
User Interface	Large selectable cards or dropdowns with plain-language descriptions; technical millimeter-per-hour values are supporting information, not the primary input method.
Required Fields	Rainfall intensity and duration are both required before susceptibility evaluation.
Stable Identifiers	The API uses stable database identifiers such as rainfall_category_id and duration_id rather than relying on display labels. This allows wording changes without breaking rules.
Unsupported Scenario	If no approved rule coverage exists for a selected combination, return Insufficient Data/Unsupported Scenario instead of guessing.
Saved Preference	The user may save a default scenario for convenience. A saved preference is not a forecast and does not imply current conditions.
Admin Control	Authorized administrators maintain category names, thresholds, descriptions, source/reference, version, allowed durations, active status, and effective date. Changes require publication of a new reference version.

Feature 5 - Dynamic and Movable Flood Susceptibility Map

The FloodSense map is not a fixed image. It is an interactive Geographic Information System (GIS) view composed of a basemap plus FloodSense-owned geographic layers. Approved barangay/susceptibility-zone polygons are stored in PostGIS and sent to the mobile application as geographic data. The same polygon can change color when the rainfall scenario changes; the geometry does not need to be recreated for every scenario.

Aspect	Detailed Specification
Mobile Mapping Technology	Flutter + flutter_map.
Basemap	OpenStreetMap-compatible raster/vector tile service used only as geographic visual context. The FloodSense susceptibility polygons and evacuation-center data are separate overlays controlled by the project.
Required User Interactions	Pan/move map; pinch or button zoom; tap polygon; tap evacuation-center marker; open/close popup; show/hide supported layers; open legend; recenter to Bacoor; enter/exit Check My Area mode; drag or reposition the user pin.
Susceptibility Polygon Layer	Each published zone has a stable zone identifier and Polygon/MultiPolygon geometry. The application receives classification results keyed by zone ID and applies style dynamically.
Color Rules	Low = green; Moderate = yellow; High = orange; Very High = red. Insufficient Data/Uncertain use a neutral gray or clearly distinct pattern and are labelled as limitation states rather than normal susceptibility classes.
Dynamic Recoloring	User changes rainfall intensity/duration -> Flutter requests a new scenario assessment set -> Django/Expert System returns classifications -> flutter_map updates each polygon fill based on class -> map stays at the same position/zoom when practical.
Zone Tap Popup	Area name; parent barangay; susceptibility class; rainfall scenario; short meaning; data/rule version; View Explanation; Check My Area or Start DSS; evacuation-resource shortcut when applicable.
Evacuation Center Layer	Displays verified evacuation centers as point markers. Markers remain geographically fixed unless the administrator edits the official record. They do not change susceptibility colors.
Layer Controls	Minimum layers: Susceptibility Zones, Barangay Boundaries if separate, Evacuation Centers. Additional layers are added only when approved data and a clear user need exist.
No Manual User Editing	Residents can interact with the map but cannot edit official polygons, move evacuation-center records, change rule outputs, or manually paint susceptibility colors.
Performance	For Bacoor-scale geometry, serve simplified GeoJSON where necessary, use spatial indexes, cache the active published map version, and avoid sending unnecessary historical/draft geometry to mobile users.
Accessibility	Legend uses text labels in addition to color. Do not rely on color alone. Tappable polygons/markers must expose textual information.

Feature 6 - Expert System Flood Susceptibility Assessment

FloodSense uses a Rule-Based Expert System. The presence of expert-provided data alone is not enough to make a system an Expert System; the expert knowledge must be formalized into a knowledge base and the system must apply an inference procedure to that knowledge. In FloodSense, authorized experts and approved institutional sources provide/validate the facts, relationships, thresholds, exceptions, and preparedness knowledge. The research team encodes the approved susceptibility knowledge into versioned rules, and the inference engine applies those rules to the current scenario.

Aspect	Detailed Specification
Expert System Type	Rule-based production system represented with decision tables and evaluated using forward chaining.
Knowledge Providers	Intended sources from the existing project documents include the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), City Planning and Development Office/other approved city offices, PAGASA references, and other authorized providers that are formally approved for the research.
Knowledge Base Content	Geographic zone profiles; rainfall categories/durations; historical flood-related attributes where approved; susceptibility decision rules; exceptions; output classes; rule priorities; source/provenance links; reviewer/approval records; versions; effective dates; explanation text; limitations.
Assessment Inputs	1) Zone/location. 2) Rainfall intensity category. 3) Rainfall duration. 4) Active zone-profile facts. 5) Active approved rule-set version. 6) Any additional fact required by the approved rule design.
Forward Chaining	Start with known facts -> load candidate approved rules -> test each required condition -> fire matching rules -> resolve applicable conflicts -> produce final class or safe limitation state -> generate explanation.
Primary Outputs	Low, Moderate, High, Very High.
Limitation Outputs	Uncertain: equally applicable approved rules conflict and cannot be safely resolved. Insufficient Data: required facts/rules are missing or the scenario is unsupported.
Explanation Output	Area/zone; scenario; result; matched rule identifier/title; summarized matched conditions; source/version; effective/update date; limitation/confidence note when available; plain-language meaning; reminder to follow official warnings.
Consistency Requirement	The same zone + same rainfall scenario + same published rule-set version must produce the same result in the map, Results Summary, Check My Area, and DSS context.
No Machine Learning	The core susceptibility classification does not train on resident behavior, infer from a statistical Machine Learning model, or change rules automatically.
No Artificial Intelligence Classification	A future Artificial Intelligence (AI) conversational feature, if ever approved, cannot produce or override susceptibility classifications.
Admin Maintenance	Authorized administrators enter/update draft rules and source metadata through the web system. Only approved/published rules can affect resident-facing outputs.

Feature 7 - Results Summary and Explainable Result Screen

Aspect	Detailed Specification
Results Summary	Scrollable/searchable list of supported geographic units under the active rainfall scenario. Each row shows area name and susceptibility class badge. Optional filters: class and barangay.
Result Screen	Area name; parent barangay; selected rainfall intensity; selected duration; class and color; short meaning; rule/explanation details; data/rule version; last update; limitation state if any; official-source reminder.
Why This Result?	Opens the explanation facility showing the matched rule in plain language rather than exposing raw code. Example structure: Because the selected area has [approved zone fact] and the selected scenario meets [approved rainfall condition], Rule [ID] classified the area as High susceptibility.
Data Provenance	Show source organization/document labels and update/version metadata that are approved for public display. Detailed internal review notes can remain admin-only.
Actions	View on Map; Check My Area; Start DSS; Find Evacuation Centers; Change Rainfall Scenario.
Stale Result	If scenario/location/map/rule version changes, the app labels the current result outdated and requires re-evaluation before it is reused.

Feature 8 - Check My Area

Aspect	Detailed Specification
Purpose	Allow the resident to manually select a location and obtain the susceptibility result for the supported zone containing that point without requiring GPS permission.
Interaction	Open map -> show movable pin -> user drags/repositions pin -> confirm selected point.
Coordinate Handling	Flutter sends the selected latitude/longitude only for the current request over HTTPS. By default, exact coordinates are not stored in user history.
Spatial Resolution	GeoDjango/PostGIS performs point-in-polygon lookup against the active published polygons and returns the most specific supported zone. The application displays the actual resolution and never claims street-level precision if the rules are only barangay/zone level.
Boundary Case	If the point lies on an ambiguous boundary, in an invalid geometry, or outside the supported study area, the app asks the user to reposition the pin or select an area manually rather than making a false precise claim.
Expert System Link	After zone resolution, the same Rule-Based Expert System evaluates the zone for the current rainfall scenario.
Output	Resolved zone/area; parent barangay; scenario; class/color; explanation; highlighted polygon; Start DSS; Find Evacuation

Aspect	Detailed Specification
	Centers.
Scenario Change	When rainfall changes, the existing pin can be re-evaluated. The resolved zone may be reused only if the active map version has not changed.
Privacy	No background GPS, no continuous tracking, no default location history, and no storage of precise pin coordinates unless a future approved requirement explicitly needs it.

Feature 9 - Decision Support System (DSS): Guided Preparedness Decision Tree

The Decision Support System (DSS) is separate from the Expert System. The Expert System answers: What is the susceptibility classification for this area under the selected scenario, and why? The DSS answers: Given that result and the user's situation, what approved pre-event preparedness information or actions should the user consider?

Aspect	Detailed Specification
DSS Type	Database-driven rule/decision-tree Decision Support System.
Trigger	Available after a valid Expert System assessment. It can also be opened from the dashboard for general preparedness topics; if a scenario-specific recommendation is requested without a valid assessment, the DSS instructs the user to run an assessment first.
Initial Context	Susceptibility class; zone/barangay; rainfall intensity; duration; published DSS version; user's current session answers.
Question Flow	Load appropriate start node -> show one simple situational question -> user chooses an allowed answer -> evaluate transition -> load next question or action group -> continue until completion.
Supported Question Areas	Go-bag readiness; important documents; family communication plan; medicines/essential needs; children/older persons/persons needing assistance; transportation readiness; awareness of official advisories; evacuation-center awareness; securing valuables; understanding susceptibility meaning.
Output	Prioritized approved preparedness actions, reason for each action, official-source reminder, evacuation-resource prompt when applicable, and completion summary.
Class-Aware Behavior	Low: awareness and basic preparation. Moderate: strengthen household readiness, secure important items, verify route/center information, prepare go-bag. High: high readiness, avoid unnecessary travel through flood-prone areas, prepare household members to follow official instructions. Very High: urgent attention to official BDRRMO/PAGASA directives and verified evacuation resources without FloodSense independently ordering evacuation.
User Control	Back to previous question when safe; restart; change answers; exit; change rainfall/location. A changed assessment invalidates scenario-dependent DSS progress.

Aspect	Detailed Specification
Deterministic Behavior	For the same published DSS version, same starting context, and same answer path, the DSS must return the same action set.
Admin Control	Administrators maintain questions, answer options, transitions, susceptibility applicability, action groups, explanations, ordering, sources, review status, and published versions.
Safety Boundary	The DSS supports decisions; it does not act as an official evacuation authority, live route-safety service, medical decision tool, or real-time flood forecast.

Feature 10 - DSS Preparedness Help and Supported User Questions

The research-team idea that the DSS should provide what the user asks is implemented in the core system as controlled preparedness help rather than an unrestricted chatbot. This keeps the feature feasible and ensures that answers come from approved preparedness content. Users can choose a help topic or search supported keywords/questions; the system retrieves the approved explanation/action associated with that topic and current assessment context.

Aspect	Detailed Specification
Purpose	Let a resident ask for a specific kind of preparedness support without completing an entire decision path when a targeted answer is sufficient.
Entry Points	Dashboard DSS card; DSS screen Help/What do you need? button; susceptibility result screen; evacuation-resource screen.
Supported Topics	What does my susceptibility level mean?; What should I put in a go-bag?; What should I prepare at home?; How do I protect documents/valuables?; What should my family plan include?; What if someone needs assistance?; Where can I find evacuation resources?; What should I monitor before a flood event?; How do I use Check My Area?
Question Method	Recommended MVP: selectable question/topic cards plus keyword search against approved topic titles/tags. This does not require a Large Language Model and is easier to validate.
Context Use	When a valid assessment exists, the help result may include class-specific guidance. Without an assessment, only general approved preparedness content is shown.
Answer Source	Versioned preparedness_action / help_topic records reviewed and published through the web administration system.
Unsupported Question	If the question/keyword does not match a supported topic, the system states that it can only provide approved FloodSense preparedness information and offers the supported categories.
Future Natural-Language Option	A conversational AI layer may be investigated later only if separately approved. It is not required for the Minimum Viable Product (MVP), Expert System, dynamic map, or DSS.

Feature 11 - Find Nearest Verified Evacuation Center

Aspect	Detailed Specification
Trigger	Available from the dashboard, map, Check My Area result, result screen, or DSS recommendation.
Data Eligibility	Only published, active, verified evacuation-center records are shown. Each record must include coordinates and a validation date; additional fields are shown only if approved and current.
Location Source	Use the current manual Check My Area pin when available. If no pin exists, the user is asked to place one or select an area. Home barangay is not treated as exact location.
Distance Query	GeoDjango/PostGIS orders centers by geographic distance from the selected point. A defined search radius may be used, then expanded when too few results exist.
Map Display	Evacuation-center marker; name; barangay/address; current published status; last verified date; estimated straight-line distance; optional capacity/contact if the responsible office provides and approves those fields.
Ranked List	Nearest verified centers first. The UI states that distance ranking does not guarantee that a route is safe or that a center is currently accepting evacuees unless its verified status specifically confirms this.
Directions	Get Directions deep-links the destination coordinate to the installed/default navigation application.
Admin Control	Add, move, edit, verify, activate/deactivate, and version evacuation-center records from the web map/form.

Feature 12 - Account and Preferences

Aspect	Detailed Specification
Profile Fields	Full/display name; email address; home barangay; optional default rainfall intensity/duration.
Email	Primary login identifier. Changing email requires a verified change flow; otherwise it is read-only from the normal profile screen.
Password	Change/reset through Django authentication. Administrators can initiate a reset or disable an account but can never read the password.
Home Barangay	Convenience preference only. May control initial map focus or default list filter.
Default Rainfall Scenario	Optional user convenience. It does not represent current weather.
Privacy Information	Explains account data, manual pin behavior, retention rules, and contact path for privacy questions.
Delete Account	Recommended public deployment function: confirmed deletion request removes/anonymizes profile data according to the approved retention policy while preserving audit records only where legally/academically justified.

Aspect	Detailed Specification
Logout	Ends authenticated session and removes local user-specific tokens/state.

Feature 13 - Degraded / Offline Behavior

Offline capability is useful but should not increase thesis risk before the core online system works. The recommended development order is online-first. After the online Expert System, DSS, map, and admin workflow are stable, the team may cache selected public data for degraded connectivity.

Aspect	Detailed Specification
Core Requirement	The Minimum Viable Product (MVP) may require internet connectivity for assessment because the authoritative rules and spatial data live on Django/PostGIS.
Recommended Later Cache	Latest published rainfall reference; disclaimer; map geometry or simplified zones; most recent scenario classification cache; DSS preparedness content; evacuation-center snapshot; data/version manifest.
Offline Warning	Always display cached data date/version and state that it may not be current.
No Fabrication	If no valid cache exists, show Service unavailable / Internet required rather than generating a result.
Synchronization	When online again, check active version manifest and replace cached public data only after the new bundle is complete and valid.

Feature 14 - Data, Methodology and Transparency

Aspect	Detailed Specification
Purpose	Make FloodSense understandable, traceable, and academically defensible.
Public Content	Data-source organizations; geographic resolution; map/data version; rainfall-reference version; Rule-Based Expert System type; simple forward-chaining explanation; susceptibility class definitions; Uncertain/Insufficient Data explanation; DSS purpose; evacuation-center verification date; map and routing limitations; privacy note; official-source reminder.
Why It Matters	The user should be able to understand that the system applies expert-approved knowledge to a selected planning scenario and that the output is not a live flood guarantee.
Admin Control	Published public methodology/disclaimer text is maintained through Django with version/effective-date records.

5. Complete Expert System Specification

5.1 Expert System Type and Role

The recommended Expert System is a rule-based production system represented by decision tables and executed through forward chaining. This type is selected because the thesis problem requires transparent use of expert-approved factual relationships rather than pattern learning from large labelled datasets. Every classification can be traced to explicit conditions and evidence metadata.

The Expert System performs susceptibility classification only. It does not perform user authentication, map rendering, free-form chatbot generation, or turn-by-turn navigation. Those are separate services that consume or display its output.

5.2 How Expert Knowledge Enters FloodSense

FloodSense is designed around explicit knowledge acquisition. Experts do not directly edit the production rule engine unless they are also formally authorized administrators. Instead, the research team collects and structures the expert/data input, records its source, converts it into proposed facts and rules, returns the encoded interpretation for review, and publishes it only after the agreed approval process. This prevents a misunderstanding in which any uploaded data automatically becomes an Expert System rule.

Stage	Detailed Knowledge-Acquisition Process
1. Identify Expert/Data Authority	Confirm which office/person is authorized to provide or validate the particular information: geographic boundary, flood history, susceptibility relationship, rainfall reference, preparedness action, or evacuation-center record.
2. Collect Source Material	Receive approved datasets, maps, historical records, interviews/elicitation forms, written procedures, or other documented evidence. Record provider, date, coverage, units, permitted use, and version.
3. Define Supported Geographic Unit	Determine whether the evidence supports barangay, sub-barangay, or custom polygon/zone output. The system must not create a finer public resolution than the evidence supports.
4. Extract Candidate Facts	Convert the source/expert statement into structured facts such as zone profile attributes, historical flood category, rainfall category relationship, exception, or limitation.
5. Build Candidate Decision Rule	Represent the expert relationship as IF conditions -> THEN susceptibility conclusion, with a plain-language explanation and source links.
6. Review with Expert/Adviser	Show the encoded rule/decision table back to the reviewer so they can confirm that the system interpretation matches the intended expert knowledge.
7. Validate	Run historical or expert-labelled test cases where available. Test thresholds, unsupported combinations, boundaries, overlapping rules, and expected limitation states.
8. Approve and Publish	Only approved rules are placed in the active published rule set. Draft/rejected rules remain unavailable to residents.
9. Maintain Version	When expert knowledge changes, create a new rule/data version rather than overwriting the historical published version in place.

Expert System Principle

FloodSense is an Expert System because authorized expert knowledge is formalized into a knowledge base and applied by a deterministic inference engine. Expert-provided data is the source of knowledge; the rule base and inference process are what make that knowledge operational and explainable.

5.3 Expert System Components

Component	Detailed Function
Knowledge Base	Stores approved rules, conditions, geographic profiles, source links, reviewer/approval metadata, rule versions, confidence/limitations, and explanation text.
Fact Builder	Constructs the current working facts from user rainfall scenario, PostGIS-resolved zone, active geographic profile, and required approved attributes.
Working Memory	Temporary in-memory structure containing only facts needed for one evaluation.
Inference Engine	Retrieves candidate rules, checks all conditions, fires valid rules, applies conflict-resolution ordering, and returns a conclusion.
Conflict Resolver	Handles multiple matching rules through geographic specificity, condition specificity, active version, documented priority, and safe uncertainty behavior.
Explanation Facility	Returns the rule and facts that produced the result in both machine-readable and plain-language form.
Version Manager	Ensures assessments use one immutable published rule-set version and records effective dates.
Rule Test Harness	Allows admins/researchers to run hypothetical facts against draft rules before publication and compare expected vs actual results.
Audit Trail	Records rule creation, edits, review, approval, publish, supersede, rollback, and actor/time.

5.4 Fact Model

- Resolved zone_id and parent barangay_id.
- Rainfall category ID and its published threshold version.
- Duration/accumulation period ID.
- Approved baseline susceptibility or zone-profile attributes if supported by source data.
- Historical flood frequency/depth category where supported.
- Environmental facts such as elevation/drainage class only when approved and encoded consistently.
- Active map/dataset version.
- Active rule-set version.
- Evaluation timestamp and application version for diagnostics; precise user pin should remain transient by default.

5.5 Rule Structure

Rule Field	Purpose
rule_id	Stable unique identifier, e.g., FS-ZONE-0001.
title	Human-readable rule name.
zone_scope	Zone, barangay, city fallback, or defined geographic scope.
rainfall_category_id	Approved rainfall category condition.
duration_condition	Exact duration or minimum/maximum range.
baseline_condition	Optional required zone-profile condition.

Rule Field	Purpose
additional_conditions	Structured list of expert-approved conditions.
output_class	Low, Moderate, High, or Very High.
priority	Documented conflict-resolution priority; never used without rationale.
confidence_or_limitation	Expert-assigned confidence/notes where the research methodology supports it.
explanation_template	Plain-language reason presented to users after variables are substituted.
source_ids	Links to evidence/provenance records.
reviewer / approval_status	Who reviewed and whether Draft, Under Review, Approved, Published, Superseded, Rejected.
version / effective_date	Controls reproducible public evaluation.

5.6 Forward-Chaining Evaluation Sequence

Order	Operation	Detailed Behavior
1	Receive assessment input	Validate rainfall category, duration, and location/zone selection.
2	Resolve geography	If coordinates were supplied, use PostGIS point-in-polygon to find the most specific active approved zone.
3	Load active versions	Select one published map/dataset version and one published rule-set version.
4	Build facts	Combine scenario and approved zone profile into working memory.
5	Find candidate rules	Retrieve active rules whose geographic scope and required fact types apply.
6	Evaluate conditions	Every required condition must match. Partial conditions do not fire a rule.
7	Collect matches	Store matched rule IDs and the condition evidence used.
8	Resolve conflicts	Apply geographic specificity, condition specificity, active/effective version, and documented priority.
9	Return classification	Return one normal class, Uncertain for unresolved conflict, or Insufficient Data for no supported rule.
10	Build explanation	Return matched rule, facts, source/version, plain-language meaning, and limitations.
11	Provide downstream context	Send classification/context to map styling, Results Summary, Check My Area, and the DSS. No other component may alter the returned susceptibility class.

5.7 Illustrative Rule - STRUCTURAL EXAMPLE ONLY

Important

The example below demonstrates the coding/data structure only. It is not an approved Bacoor flood rule and must not be used for public classification until expert/data validation is completed.

Aspect	Detailed Specification
Rule ID	FS-Z-EXAMPLE-001
IF	Zone = Example Zone; baseline = Moderate; rainfall category = Heavy; duration $\geq$ 6 hours; historical flood evidence category = Frequent.
THEN	High susceptibility.
Explanation	The scenario matches the approved prolonged-heavy-rain condition associated with frequent historical flooding for this example zone.
Approval	NOT APPROVED - STRUCTURAL EXAMPLE ONLY.

5.8 Expert System Use Across the Whole Application

System Feature	How the Expert System Is Used
Dashboard	Provides the primary Assess Flood Susceptibility entry point and shows whether a current valid result exists.
Rainfall Input	Creates scenario facts used by the inference engine.
Dynamic Map	Evaluates or retrieves deterministic results for every visible/published zone and converts the class to polygon styling.
Results Summary	Displays the same classifications used by the map.
Check My Area	Resolves the user pin to a zone and evaluates the rule set for that zone/scenario.
Result Explanation	Shows the rule/explanation metadata generated by the explanation facility.
DSS	Consumes the classification and context as inputs but cannot change it.
Evacuation Resources	May be emphasized based on the class, but the center data and routing remain separate from the Expert System classification.
DSS Preparedness Help	Uses the Expert System result only as read-only context when presenting approved susceptibility explanations and preparedness topics. It is not an inference engine.
Admin Web	Creates, tests, reviews, versions, publishes, supersedes, and audits the Expert System knowledge base.

6. Complete Decision Support System (DSS) Specification

6.1 DSS Role

The DSS is a deterministic pre-event preparedness guidance system. It begins after or alongside an Expert System assessment. The Expert System answers what the area's susceptibility classification is under the selected scenario; the DSS answers what approved preparation steps are relevant to the user's situation. This separation prevents the preparedness logic from being confused with the susceptibility classifier.

6.2 DSS Data Model

Entity	Key Data
dss_versions	Version, title, status, effective date, reviewer/approval metadata.
dss_nodes	Node ID, type (question/action/info/end), text, susceptibility applicability, geographic applicability if any, order/metadata.
dss_options	Allowed answer values and labels for a question.
dss_transitions	From node, option/condition, to node, priority.
preparedness_actions	Action title, detailed instructions, category, susceptibility applicability, source, official reminder, approval status.
node_actions	Links one or more preparedness actions to a node/end state.
dss_sources	Evidence and approval records for questions/actions.
dss_sessions	Prefer transient/local session state. If analytics are approved, store only de-identified completion metrics rather than sensitive household answers.

6.3 Detailed Decision Flow

Stage	Behavior
Context Check	Verify that assessment_result is current for the selected rainfall scenario and location.
Start Node Selection	Choose the active DSS start node for the susceptibility class and any approved context conditions.
Question Presentation	Show one clear question and only the configured answer options.
Branch Evaluation	Map the selected option plus context to one transition. If configuration is invalid or ambiguous, stop safely and show general approved guidance instead of guessing.
Action Retrieval	When an action node is reached, return the approved actions and source/update metadata.
Evacuation Resource Link	If the branch recommends reviewing evacuation resources, show the nearest verified-center action without making a route-safety claim.
Completion	Summarize prioritized actions, official-source reminder, and options to review answers, restart, or return to map.
Scenario/Location Change	Invalidate the session and require restart because its context changed.

6.4 Example DSS Path - ILLUSTRATIVE ONLY

Step	Example
Context	Expert System result = High; scenario = Heavy / 6 hours; area = Example Zone.
Question 1	Do you have a prepared go-bag? -> No.
Action	Show approved essential go-bag preparation checklist.
Question 2	Does anyone in your household need mobility, medication,

Step	Example
	infant, elderly, or other special assistance? -> Yes.
Action	Show approved assistance-planning guidance and remind user to coordinate early with family/barangay/official channels.
Question 3	Do you know the verified evacuation centers for your area? -> No.
Action	Open Evacuation Resources with the user's pin and verified center list.
Completion	Show summary and reminder to follow current BDRRMO/PAGASA/barangay instructions.

6.5 DSS Features Across the System

- Dashboard tile shows Start/Resume Preparedness Guidance.
- Result screen presents Start DSS after classification.
- Dynamic map zone popup can open the DSS after a zone assessment.
- Check My Area passes the pinned zone and current class into the DSS.
- Evacuation center lookup is linked as a DSS action rather than being treated as a classification result.
- The DSS Preparedness Help feature can explain an approved action but cannot alter the deterministic branch unless the user intentionally answers a different decision-tree question.
- Admin web provides the DSS builder, versioning, preview/test, and publish controls.

6.6 How the DSS Provides What the User Asks

The core DSS supports user-requested preparedness information through two controlled interaction styles: (1) guided decision-tree questions, and (2) direct supported-help topics/keyword search. Both styles use the same approved preparedness knowledge base. This gives the user useful answers without requiring an unrestricted conversational model.

User Need	DSS Handling
I want to know what I should prepare	Open the preparedness branch or help topic appropriate to the current susceptibility class and show the approved action list.
What should be in my go-bag?	Retrieve the approved go-bag preparedness action/topic and, if applicable, add class-specific priority wording.
What does High/Very High mean?	Retrieve the approved susceptibility explanation plus the current Expert System result metadata. The DSS does not reclassify the area.
What should my family do?	Open family communication/household readiness questions and return approved actions based on the answers.
Someone in the household needs assistance	Open the vulnerable-household assistance branch and present approved pre-event preparation guidance.
Where can I evacuate?	Open the verified evacuation-center module using the current manual pin/area context.
What should I do before leaving home?	Show the approved pre-departure preparedness action group if it is part of the validated knowledge base.
Question not covered by the knowledge base	State that FloodSense cannot answer that request from approved content and show the list of supported topics or official sources.

For implementation, each help topic should have a stable topic_id, title, keywords/tags, general response/action links, optional susceptibility applicability, source, review status, version, and effective date.

Keyword matching should be simple and transparent. A direct natural-language chatbot is intentionally deferred so that the core DSS remains testable and expert-reviewable.

7. Dynamic GIS Map and Spatial Data Specification

7.1 Geographic Data Model

Spatial Entity	Geometry	Purpose
Barangay	Polygon / MultiPolygon	Administrative context and parent geography.
Susceptibility Zone	Polygon / MultiPolygon	Most specific unit supported by approved evidence and Expert System rules.
Evacuation Center	Point	Verified center location used for map markers and proximity ranking.
Optional Reference Feature	Point/Line/Polygon	Roads, waterways, landmarks, or other context only when an approved need and data source exist.

7.2 Admin Manipulable Map

The administrator receives editing tools that mobile users never see. Geographic editing is performed in the web application using Leaflet and Leaflet-Geoman. All changes are drafts until validated and published.

- Draw polygon or multipolygon susceptibility zones.
- Edit polygon vertices; drag, rotate/cut/split only if enabled and supported by the approved workflow.
- Delete or archive draft geometry.
- Add/move evacuation-center point markers.
- Import GeoJSON and validate coordinate reference system before acceptance.
- Assign zone ID, name, parent barangay, source dataset/version, geographic resolution, notes, and active status.
- Detect invalid/self-intersecting polygons and reject publication until corrected.
- Preview susceptibility colors using a selected draft rule version and rainfall scenario before publishing.
- Compare new geometry with currently published version and record change reason.
- Publish a complete map/dataset version atomically so mobile users never receive a half-edited map.

7.3 Dynamic Scenario Coloring

Zone geometry and susceptibility result are deliberately separated. A polygon is a geographic object; its current color is a scenario-dependent presentation. This prevents the team from maintaining a separate image for every rainfall combination.

Data	Example
Geometry	zone_id = TAL-Z03; polygon stored in PostGIS SRID 4326.
Scenario	rainfall_category = Heavy; duration = 6 hours.
Expert Result	zone_id TAL-Z03 -> High.
Client Style	High -> orange fill, defined by the published legend.
Scenario Change	Torrential + 12 hours -> backend re-evaluates -> result may become Very High -> same polygon is restyled red.

7.4 Spatial Queries Required

Query	Purpose
ST_Contains / ST_Covers	Resolve a user pin to an approved zone.
ST_Intersects	Validate overlapping/related geographic features and admin edits where needed.
ST_Distance on geography	Rank verified evacuation centers by straight-line/geodesic distance.
ST_DWithin	Retrieve centers/features within a configured radius.
Bounding-box/indexed retrieval	Load only relevant map features for a viewport or reduce payload size.
Geometry validation functions	Detect invalid polygons before publish.

7.5 Basemap and Tile Policy

The map library and the basemap provider are separate. flutter_map and Leaflet display map tiles/layers; a production tile provider supplies the basemap according to its usage terms. Development may use a suitable test source, but public deployment should use an approved provider or self-hosted tiles rather than assuming unrestricted use of a public community tile endpoint.

7.6 Complete Map Object and Editing Capabilities

Object / Capability	Resident Android App	Authorized Admin Web
Pan / move map	Yes	Yes
Zoom in/out	Yes	Yes
Tap/select zone	Yes - read only	Yes - inspect/edit when in draft editor
Add susceptibility polygon	No	Yes - draw new draft polygon/multipolygon
Edit polygon vertices	No	Yes - move/add/remove vertices in draft data
Move entire polygon	No	Yes when the editing tool/workflow allows and the resulting geometry passes validation
Delete polygon	No	Draft delete/archive; published historical versions remain auditable
Import GeoJSON	No	Yes - through validated admin import
Add location/point	User may place only a temporary Check My Area pin	Yes - add authorized point records such as evacuation centers
Move evacuation-center marker	No	Yes - drag/edit coordinate then save draft/verified record
Delete evacuation center	No	Deactivate/archive preferred over destructive deletion when previously published
Change susceptibility color manually	No	No for public classification; color is derived from Expert System output. Admin may maintain the legend/style configuration only.
Show/hide map layers	Yes for supported public layers	Yes for editing/reference layers
Popup information	Yes - public details	Yes - full editable metadata plus validation/source/status
Nearest-center search	Yes	Preview/test available
Draw arbitrary route	Not a core feature	Not a core feature; external navigation handles routing

7.7 Map Publication Rule

The administrator edits Draft geographic objects. Draft geometry is never sent to residents. Publication creates or activates a complete map version. Mobile clients request only the active published version. This prevents a resident from seeing a polygon while it is half-edited. If an administrator discovers a critical geographic error after publication, the system can publish a corrected version or roll back to a previously approved version according to the recorded change-control process.

8. Complete Web Administration System

The Admin Web System is the controlled maintenance interface for FloodSense. It is not exposed to normal residents. Admin capabilities are intentionally broader than mobile capabilities because the admin maintains the information that drives the public system.

8.1 Admin Authentication and Authorization

Aspect	Detailed Specification
Login Credentials	Admin email + password. Account must already be assigned an authorized admin role in server-side authorization data.
Additional Security	Optional later enhancement: add Multi-Factor Authentication (MFA) if required by the institution. It is not necessary to implement the current thesis features.
Role Enforcement	Django checks the authenticated admin session and server-side <code>is_staff/group/model</code> permissions before protected views or changes are allowed. Hiding a button in the web page is not sufficient security.
Session Security	Django server-side session authentication protects the web admin. Logout ends the session; disabling/removing staff permissions blocks future privileged actions. Session timeout can be configured for production.
Password Visibility	No administrator can read any user or admin password.
Audit	Successful/failed privileged logins and sensitive changes should be logged without recording passwords or full auth tokens.

8.2 Admin Dashboard

Aspect	Detailed Specification
Summary Cards	Published map version; active Expert System rule version; active DSS version; rainfall-reference version; number of active zones; number of verified evacuation centers; last publish date; pending drafts/reviews.
Warnings	Invalid geometries; rules with missing sources; draft changes not reviewed; evacuation centers past verification date; unpublished DSS branches; failed sync/deployment; API health issues.
Recent Activity	Latest rule/map/DSS/content changes with administrator, timestamp, and status.
Quick Actions	Add evacuation center, open map editor, create rule draft,

Aspect	Detailed Specification
	open DSS builder, import dataset, publish approved version, view audit logs.
No Public Analytics by Default	Do not collect or expose sensitive location histories merely to populate the dashboard. Usage analytics must be separately justified and privacy-reviewed.

8.3 Geographic / Map Data Management

Aspect	Detailed Specification
Map Editor	Interactive Bacoor map with draw/edit/delete tools and layers for barangays, susceptibility zones, and evacuation centers.
Zone Form	Zone ID, name, parent barangay, geometry, source dataset/version, resolution, notes, active/draft status.
Import	Upload GeoJSON; validate structure, coordinate system, duplicate IDs, missing attributes, invalid polygons, and overlap rules before saving as draft.
Versioning	Every publish creates a map/dataset version. Previous published geometry remains available for audit/rollback.
Preview	Select draft rule version + rainfall scenario and preview zone colors before publication.
Publish Safety	Block publish if required geometry validation fails or mandatory source metadata is missing.

8.4 Expert System Knowledge Base Management

Aspect	Detailed Specification
Rule List	Search/filter by status, zone, rainfall category, output class, version, reviewer, and source.
Create Rule	Select geographic scope; add conditions using structured fields/operators; select conclusion; add explanation; assign source records; set rationale/priority; save as Draft.
Edit Rule	Edit only draft/review versions. Published rules are immutable; changes create a new version.
Rule Condition Builder	Structured fields such as rainfall category, duration range, baseline category, historical evidence class, and other approved facts. Avoid free-form executable code entered by admins.
Rule Test Sandbox	Enter hypothetical zone/scenario/facts -> run draft engine -> show matched rules, fired conditions, result, explanation, and conflict details. Results are labelled TEST ONLY.
Batch Test	Run all approved/historical test cases against a draft rule set and export comparison results.
Review Status	Draft -> Under Review -> Approved -> Published; or Rejected/Superseded. Exact approval roles may be simplified in MVP but the status history must remain.
Publish	Only approved complete rule versions can become active.

Aspect	Detailed Specification
	Publication records who published, when, and why.
Rollback	Restore a previously published version when a critical defect is found; record rollback reason.
Source Link	Each public rule must link to one or more provenance records and expert review information.

8.5 DSS Builder and Preparedness Content Management

Aspect	Detailed Specification
Visual Tree	Canvas/tree view of question, information, action, and end nodes with connections.
Question Editor	Question text; help text; susceptibility applicability; optional geographic applicability; answer type; allowed options.
Branch Editor	For each option, choose the next node and optional condition/priority. System validates unreachable nodes, loops, missing transitions, and dead ends.
Action Editor	Preparedness action title, detailed steps, category, susceptibility applicability, source, official reminder, display priority.
Preview Mode	Simulate the mobile DSS from a selected susceptibility result and sample answers before publishing.
Versioning	Draft and published DSS versions are immutable once active; edits create new draft version.
Safety Review	Flag language that implies real-time road safety, unsupported certainty, or independent evacuation orders for mandatory review.
Offline Package	Publishing generates a normalized DSS data bundle that can be cached by the mobile app.

8.6 Rainfall Parameter Management

Aspect	Detailed Specification
Fields	Category ID, display name, description, lower/upper threshold or reference text, unit, allowed durations, source, effective date, version, active status.
Change Control	Published threshold changes create a new version because they can affect every Expert System rule.
Dependency Check	Before publishing a changed category, show affected rules/scenarios and block publication when referenced categories would become invalid.

8.7 Evacuation Center Management

Aspect	Detailed Specification
Add	Click Add Center -> click map or enter coordinates -> complete center form -> validate -> save Draft.
Fields	Center ID; name; barangay; full address; latitude/longitude; operational/public status; capacity if approved; contact if approved; responsible office; source; validation date; notes.
Map Editing	Drag marker to correct position; verify coordinate preview before save.
Verification	A center cannot be public unless marked Verified with responsible source and validation date.
Deactivate	Do not hard-delete a previously published center when historical audit is needed; mark inactive/superseded with reason.
Public Output	Only active verified records are sent to normal users.

8.8 Data Source and Provenance Management

Aspect	Detailed Specification
Purpose	Document where geographic profiles, historical flood evidence, rainfall references, preparedness content, and evacuation-center records came from.
Fields	source_id; provider/office; dataset/document name; request/reference number if available; date received; geographic coverage; temporal coverage; units; license/permitted use; contact/owner; validation status; update date; file attachment link; notes.
Relationships	Rules, zones, preparedness actions, rainfall categories, and evacuation centers can reference one or more source IDs.
Public Visibility	Only appropriate non-sensitive source metadata is exposed in the mobile transparency screen.

8.9 Content Management

Aspect	Detailed Specification
Managed Content	Onboarding, disclaimer, susceptibility definitions, legend descriptions, methodology summary, help text, official reminders, contact information, privacy notice link, FAQ.
Versioning	Published content has version/effective date. Material disclaimer changes can trigger re-acknowledgement.
Preview	Admin previews mobile-formatted content before publish.

8.10 User Account Management

Aspect	Detailed Specification
Allowed Admin Actions	Search by email/account ID when support is required; view account status and basic profile; disable/reenable account if policy permits; view verification status; initiate support workflow.
Prohibited Admin Actions	Read passwords; impersonate a user without an approved audited support mechanism; view unnecessary location histories; silently change security credentials.
Profile Editing	Prefer allowing users to edit their own full/display name and home barangay. Admin overrides should be rare and audited.

8.11 Audit Log and Change History

Aspect	Detailed Specification
Events	Admin login/security events; create/edit/delete/archive drafts; rule review/publish/rollback; DSS publish; map import/edit/publish; evacuation-center verify/deactivate; content changes; user disable actions; configuration changes.
Fields	event_id, timestamp, actor_id, action, entity_type, entity_id, version, before/after summary or change record, reason/comment, request IP/device metadata where legally appropriate.
Immutability	Audit records should not be editable through normal admin screens.
Retention	Set a documented retention policy appropriate to research/system governance.

8.12 Backup, Export and Recovery

Aspect	Detailed Specification
Database Backup	Use managed provider backup features according to the selected plan and keep periodic logical exports before major publications.
Knowledge Export	Export rule set, DSS version, zone metadata, rainfall parameters, evacuation centers, and source register as JSON/CSV/GeoJSON for thesis evidence and disaster recovery.
Restore	Test restore procedure in staging; never test recovery for the first time during a production failure.
Rollback	Map/rule/DSS/content version rollback is separate from full database disaster recovery and should be available from controlled admin workflows.

9. Backend, Application Programming Interface and Database Design

9.1 Django Backend Responsibilities

- Create and manage the custom email-based user model, resident profiles, administrator roles, password validation, password reset, and account status.
- Authenticate mobile Application Programming Interface (API) requests through Django REST Framework (DRF) and enforce permissions before protected processing occurs.
- Protect the web administration system with Django session authentication, Cross-Site Request Forgery (CSRF) protection, and server-side staff/permission checks.
- Validate every mobile/admin input again on the server even when Flutter or JavaScript already performed client-side validation.
- Use GeoDjango and PostGIS to resolve points to zones, retrieve published polygons, validate geographic data, and calculate evacuation-center proximity.
- Run the Rule-Based Expert System, including fact construction, rule matching, forward chaining, conflict resolution, limitation states, and explanation generation.
- Run the database-driven Decision Support System (DSS), including node selection, answer transitions, preparedness-action retrieval, and supported help-topic lookup.
- Return published GeoJSON and scenario-specific zone classifications for the dynamic mobile map.
- Manage Django Admin and custom administrator pages for map editing, Expert System rules, DSS content, rainfall parameters, evacuation centers, source metadata, public content, review/version states, and audit logs.
- Keep draft/unapproved records unavailable to normal mobile users.
- Create reproducible version identifiers so map, rule set, rainfall reference, DSS, and public content are not silently mixed across releases.
- Record privileged administrative changes in an audit history.
- Provide a health/status endpoint and controlled error responses for mobile diagnostics without exposing stack traces or secrets.

9.2 Recommended Mobile API Groups

API Group	Example Responsibilities
/api/v1/auth/register	Create a resident account after server-side field validation.
/api/v1/auth/login	Verify email + password and return the selected mobile authentication token/session representation.
/api/v1/auth/refresh	Refresh an expiring mobile access token when JSON Web Token (JWT) authentication is used.
/api/v1/auth/logout	Revoke/blacklist the mobile session when supported and clear server-side session state where applicable.
/api/v1/auth/password-reset	Request password reset and complete the reset using a time-limited token/link workflow.
/api/v1/profile	Read/update allowed resident profile fields such as full/display name, home barangay, and default rainfall scenario.
/api/v1/public/config	Published rainfall categories, durations, susceptibility legend, version manifest, disclaimer/methodology metadata.
/api/v1/map/zones	Return active published zones and GeoJSON geometry/metadata. May support viewport/bounding-box filtering if needed for performance.
/api/v1/assessments	Evaluate a selected zone or coordinate under a rainfall scenario and return classification, explanation, source/version metadata, and limitation state.

API Group	Example Responsibilities
/api/v1/assessments/map	Evaluate/return the classification set used to color all active map zones for one rainfall scenario.
/api/v1/locations/resolve	Resolve a manually selected point to the most specific supported published zone.
/api/v1/dss/start	Start a DSS session from a valid assessment context or general-preparedness mode.
/api/v1/dss/answer	Submit the selected answer and return the next question/action set.
/api/v1/dss/help	Retrieve approved preparedness help topics/keyword matches and context-aware approved answers.
/api/v1/evacuation-centers	Return verified active centers; optionally rank by distance from a supplied manual pin.
/api/v1/version	Return current public map/rule/rainfall/DSS/content versions for synchronization and stale-result detection.

Administrator maintenance may use normal Django forms/views instead of exposing every administrative action as a public REST endpoint. The custom map editor can use small authenticated JSON endpoints under a protected /manage/ namespace where asynchronous map save/validation is useful. This keeps the public mobile API small and reduces unnecessary attack surface.

9.3 Initial Database Entities

Entity / Table	Purpose and Important Fields
users	Custom Django user model: user_id, email (unique/login), password hash, is_active, is_staff/admin role flags, created_at, last_login.
user_profiles	Full/display name, home_barangay_id, optional default rainfall category/duration, privacy/disclaimer acknowledgement version.
barangays	Barangay identifier, name, published geometry when needed, source/version metadata.
susceptibility_zones	Zone identifier, name, parent barangay, PostGIS Polygon/MultiPolygon geometry, geographic-resolution metadata, source/version, status.
map_versions	Version identifier, status, effective date, created/reviewed/published by, notes; links a coherent set of geographic objects.
rainfall_categories	Stable category identifier, display label, threshold/reference fields, description, active flag.
rainfall_versions	Reference version, source, effective date, status, notes.
rainfall_durations	Stable supported duration options and active/version relationships.
zone_profiles	Expert/data-approved static facts for each zone that may be consumed by the rule engine.
expert_rule_sets	Rule-set version, status, effective date, reviewer/publisher metadata.
expert_rules	Rule ID/title, geographic scope, output class, priority, explanation template, source links, status/version.
expert_rule_conditions	Structured field/operator/value conditions belonging to a rule. Avoid administrator-entered executable Python.
expert_reviews	Review/approval records, reviewer, date, decision, notes, linked rule/rule-set or data version.

Entity / Table	Purpose and Important Fields
preparedness_actions	Approved action title, instructions, category, applicability, source, review/version/status.
dss_versions	Published/draft Decision Support System version metadata.
dss_nodes	Question, action, information, or end node; text; applicability; version.
dss_options	Allowed answers for question nodes.
dss_transitions	From-node + answer/condition -> to-node; ordering/priority where required.
dss_help_topics	Supported preparedness questions, keywords/tags, approved response/action links, applicability, version.
evacuation_centers	Center ID, name, PostGIS Point geometry, barangay/address, status, validation date, responsible source, optional approved capacity/contact.
data_sources	Provider/office, dataset/document name, reference number, date received, geographic/temporal coverage, use permission, update date, validation status, notes.
content_versions	Onboarding, disclaimer, methodology, legend/help text, effective date and status.
audit_logs	Actor, timestamp, action, entity type/ID, version, before/after summary or structured change reference, reason when required.
system_settings	Non-secret configuration such as currently published version identifiers. Production secrets are environment variables, not database configuration rows.

9.4 GeoDjango and PostGIS Spatial Behavior

Spatial Need	Recommended Implementation
Store zones	GeoDjango PolygonField/MultiPolygonField backed by PostGIS geometry.
Store evacuation centers	GeoDjango PointField backed by PostGIS.
Check My Area	Use a point-in-polygon query such as contains/covers/within logic against the active published zone geometry. Boundary behavior must be explicitly tested.
Nearest centers	Use PostGIS geography/distance operations and spatial indexes; rank verified centers by distance from the user-selected point.
Search radius	Use a distance-within query when a radius is desired, then order the returned centers by distance.
Geometry validation	Validate imported/drawn geometry before publication; invalid geometry blocks publish.
Coordinate system	Use a documented Spatial Reference System Identifier (SRID). Geographic exchange may use World Geodetic System 1984 / EPSG:4326; distance calculations should use geography or an appropriate projected system.
Spatial indexes	Create spatial indexes on zone and center geometry fields so point containment and distance queries remain efficient.

9.5 Data Version Consistency

Every public assessment must be reproducible. The response should identify the active map/dataset version, Expert System rule-set version, rainfall-reference version, DSS version when relevant, and evaluation timestamp. A mobile screen must never combine a zone polygon from one published map version with a susceptibility result generated for a different incompatible version. If the server publishes a new version while a user is viewing an old result, that result is marked stale before it is reused.

10. Security, Privacy and Safety Controls

Control Area	Specification
Transport	All production mobile API and web-administration traffic uses Hypertext Transfer Protocol Secure (HTTPS).
Password Storage	Django password hashing/validators are used. Plaintext passwords are never stored, displayed, emailed, returned through the API, or written to logs.
Mobile Authentication	Email + password authenticates through DRF. Production should use an expiring/revocable token design. Tokens are stored using an appropriate secure storage mechanism on Android and transmitted only through HTTPS.
Admin Authentication	Django session authentication for the web administrator. Administrator accounts are created/authorized server-side and require <code>is_staff</code> /permission checks. Multi-factor authentication may be added later if the team or institution requires it; it is not required to build the current thesis features.
Authorization	Every privileged action is checked server-side. Hidden UI controls are convenience only and are never the security boundary.
Cross-Site Request Forgery	Django Cross-Site Request Forgery (CSRF) protection remains enabled for web administration forms and state-changing browser requests.
Database Access	Flutter and browser clients never receive PostgreSQL credentials. Only the Django server connects to PostgreSQL/PostGIS.
Secrets	Django <code>SECRET_KEY</code> , database URL/password, email credentials, and other production secrets are stored in Render environment variables, not committed to Git.
Input Validation	Django/DRF serializers and forms validate field types, lengths, allowed values, foreign-key relationships, role restrictions, and business rules. GeoJSON uploads are size/type/structure checked before parsing.
Geometry Safety	Invalid or unreviewed polygons cannot be published. Point-in-polygon logic and boundary cases are tested.
Location Privacy	Check My Area uses a manual pin. Exact coordinates are processed for the current request and are not stored in user history by default. No background tracking is required.
Audit	Privileged create/edit/review/publish/rollback/deactivate actions are recorded. Logs exclude passwords, full access tokens, and unnecessary precise user locations.
Login/Request Abuse	Use DRF throttling and/or hosting-layer rate limits for repeated authentication attempts and unusually high assessment/API traffic. Exact thresholds are selected during testing.
Safety Wording	Susceptibility is not described as a live flood forecast, current road safety, guaranteed flooding, or an official evacuation order.

Control Area	Specification
Uncertainty	Missing rule coverage, conflicting rules, invalid locations, stale versions, and outdated evacuation-center records appear as visible limitation states rather than being hidden.

11. Development-to-Deployment Plan

11.1 Recommended Project Structure

Folder / Repository	Contents
/mobile	Flutter Android project: screens, routing, API client, map layers, user-pin logic, result/DSS UI, tests, Android release configuration.
/server	Django project: authentication, Django Admin/custom management pages, Django REST Framework API, GeoDjango models, Expert System, DSS, database migrations, templates/static files, tests, deployment files.
/server/research_imports	Development-only scripts/templates for importing approved CSV/GeoJSON or constructing versioned rule data. Do not place restricted institutional source files in a public repository.
/docs	Architecture notes, API contracts, data dictionary, rule/DSS schema, deployment checklist, validation/test evidence.

A single GitHub repository containing /mobile and /server is recommended for the three-person thesis team because it reduces repository administration and keeps coordinated mobile/server changes in one review history. Separate repositories can be used later if the project grows.

11.2 Development Environments

Environment	Purpose
Local Development	Developer devices. Flutter runs against a local or shared development Django server/database. PostgreSQL/PostGIS may be installed locally or provided through a development Render database. GeoDjango native library requirements must be handled consistently across team machines.
Staging / Thesis Demonstration	Production-like Django + PostgreSQL/PostGIS environment containing approved test data or first real reviewed data. Used for adviser/expert review, user testing, and deployment verification.
Production / Pilot	Only published approved map/rule/DSS/content versions, restricted admin access, production secrets, database backup settings, and signed Android build.

11.3 Why Docker Is Recommended for the Django Server Deployment

Docker is not an application framework and is not required to write FloodSense. It is recommended for deployment because GeoDjango commonly depends on native geographic libraries such as GEOS, PROJ, and GDAL in addition to PostgreSQL/PostGIS. A small Docker image lets the team declare those operating-system dependencies once and run the same server package on staging and production. If the chosen Render native environment is proven to provide every required GeoDjango library reliably, Docker can be omitted; otherwise Docker reduces deployment surprises.

11.4 Production Deployment Sequence

Order	Deployment Task
1	Create the production Render PostgreSQL database and enable the PostGIS extension. Create a separate staging database if budget permits.
2	Configure the Django database connection through an environment variable and apply Django migrations.
3	Create the Render Django Web Service. Deploy the Django backend, DRF API, and web administration system together. Configure SECRET_KEY, DEBUG=False, allowed hosts, database URL, email settings, and any map-provider configuration as environment variables.
4	Install/configure GeoDjango native dependencies in the deployment image/runtime; verify a real PostGIS geometry query before continuing.
5	Collect/serve Django static files. Use WhiteNoise or the deployment approach supported by the selected Render configuration.
6	Create the first authorized Django superuser/admin account through the secured server shell, then create narrower administrator permissions for routine use if required.
7	Configure the basemap tile service used by Flutter and Leaflet and confirm that its production usage terms cover the expected thesis/pilot traffic.
8	Run an end-to-end staging publication test: admin creates/edits draft geographic data -> validates -> publishes -> mobile retrieves geometry -> Expert System classifies -> colors update -> Check My Area resolves -> DSS responds -> evacuation centers rank.
9	Build the Flutter Android App Bundle (AAB), configure the unique application identifier, signing/upload key, production API base URL, and release version.
10	Upload the AAB to Google Play internal or closed testing. Test on multiple Android devices and network conditions before production release.
11	After adviser/expert/system validation, publish the approved Android release and keep the previous stable server/data versions available for rollback.

11.5 Deployment Operations Kept Intentionally Simple

- Render can connect directly to the Git repository and automatically redeploy the Django service after an approved push/merge; a separate Vercel/Next.js deployment is not needed.
- Render PostgreSQL supports PostGIS, so a separate Supabase database is not required.
- No Redis/Celery background-worker stack is required for the current synchronous assessment, DSS, map, and admin workflows. Add background jobs only if later tasks are too slow for normal requests.
- No Kubernetes or microservice architecture is required. One Django service plus one managed PostgreSQL/PostGIS database is sufficient for the thesis/pilot.
- No separate object-storage service is required for the core. GeoJSON import files can be uploaded temporarily, validated, converted to database geometry, and discarded after successful import. Persistent evidence attachments can be added later only if required.
- Use Render logs and the Django health endpoint first. Add third-party error/uptime services only if operational experience shows a need.

11.6 Source Control and Release Discipline

- Use Git feature branches and pull requests for code changes.
- Never commit production secrets, database passwords, Android signing keys, or restricted institutional datasets.
- Run Flutter tests, Django tests, Expert System rule tests, DSS path tests, and migration checks before merging a release.
- Code deployment and knowledge/data publication are different operations. A new Django code version does not automatically publish draft rules or map data; administrator publication remains versioned/audited inside FloodSense.
- Tag major thesis demonstration/defense versions so the exact code used for evaluation can be reproduced.

12. Detailed Administration Workflows

12.1 Publish a New Geographic Zone Version

Step	Admin Action / System Check
1	Open Map Management and create a new Draft map version.
2	Import or draw/edit zones; assign IDs, parent barangays, source metadata.
3	Run geometry validation: valid polygon, no unintended self-intersections, required fields, source present.
4	Preview differences against current published version.
5	Test point-in-polygon for boundary samples and representative points.
6	Preview Expert System colors using a compatible draft/active rule version.
7	Submit for review/approval according to research workflow.
8	Publish atomically; version manifest changes; mobile clients receive new version on sync.
9	Monitor error logs and keep rollback option to previous published version.

12.2 Publish a New Expert Rule Version

Step	Admin Action / System Check
1	Create draft rule set from current published version.
2	Add/edit structured rules and source links; do not edit published rules in place.
3	Run rule lint checks for missing conditions, missing outputs, invalid category references, duplicate/overlapping rules, and orphaned zone references.
4	Run single-case sandbox tests and batch validation cases.
5	Review conflicts and ensure expected Uncertain/Insufficient Data behavior.
6	Record expert/reviewer approval.
7	Publish new rule-set version and effective date.

Step	Admin Action / System Check
8	Invalidate/rebuild deterministic scenario caches if used.
9	Verify mobile map/result consistency in staging/production smoke test.

12.3 Publish a New DSS Version

Step	Admin Action / System Check
1	Create draft DSS version.
2	Edit questions, options, transitions, and actions.
3	Run graph validation for unreachable nodes, cycles not explicitly allowed, missing branches, and end nodes without actions.
4	Preview all susceptibility-level start paths.
5	Review language for safety and official-authority boundaries.
6	Attach sources/approval metadata.
7	Publish and generate offline cache bundle.
8	Mobile syncs new version; in-progress sessions tied to old version may complete locally or restart according to version policy.

12.4 Maintain Evacuation Centers

Step	Admin Action / System Check
1	Receive/confirm responsible-office record.
2	Add or update center point and metadata.
3	Verify map position against source coordinates.
4	Record status, responsible source, and validation date.
5	Publish/activate center.
6	Periodically revalidate. If outdated/closed, deactivate rather than delete historical record.

13. Feature-Level Development Breakdown

The estimates below are planning ranges rather than promises. They assume three team members can work partly in parallel and that authoritative data/rule approval occurs on time. The development order should protect the thesis core: data model -> admin/map maintenance -> Expert System -> mobile assessment/map -> DSS -> evacuation resources -> deployment and validation.

Work Package	Complexity	Indicative Effort	Main Dependency / Completion Evidence
Project scaffold and environments	Low-Medium	1 week	Git repository; Flutter project; Django project; development PostgreSQL/PostGIS;

Work Package	Complexity	Indicative Effort	Main Dependency / Completion Evidence
			environment variables; basic health page.
Email/password authentication	Medium	1 week	Custom Django user model; registration/login/reset endpoints; mobile screens; admin session; role enforcement; tests.
Rainfall reference management	Low-Medium	0.5-1 week	Versioned categories/durations, admin forms, published config API, verified wording/source.
PostGIS geographic schema	Medium	1 week	Barangays/zones/centers as spatial fields; SRID decision; spatial indexes; sample geometry query.
Admin dynamic map editor	Medium-High	2-3 weeks	Leaflet + Leaflet-Geoman page; draw/edit/move/delete/import; validation; draft/publish behavior.
Mobile dynamic map	Medium-High	2 weeks	flutter_map; published GeoJSON; pan/zoom/tap; layer controls; class-based coloring; evacuation markers.
Expert knowledge model and rule tables	High	1-2 weeks after knowledge format agreed	Rule-set/rule/condition/source/ review models; admin structured rule form; no executable admin code.
Expert System inference engine	High	2-3 weeks	Forward chaining; candidate selection; conflict resolution; limitation states; explanation output; automated tests.
Expert System admin test sandbox	Medium-High	1-2 weeks	Run draft hypothetical cases; show matched/fired rules; compare expected result; no public effect.
Results Summary / explanation	Medium	1 week	Consistent map/list/result output with version metadata and stale-state handling.
Check My Area	Medium	1 week	Movable pin; point-in-polygon API; boundary/outside handling; privacy behavior; same Expert System output.
DSS data model and engine	High	1.5-2 weeks	Nodes/options/transitions/ actions; deterministic path evaluator; assessment-context checks; tests.
DSS admin builder/preview	High	2 weeks	Question/action forms; transition editing; graph validation; preview path; publish version.
DSS preparedness help	Medium	0.5-1 week	Help topics/tags, keyword lookup, class-aware approved answers, unsupported-topic fallback.
Evacuation-center management	Medium	1 week	Admin map point editor; verification fields; active/published rules.
Nearest evacuation-center mobile feature	Medium	1 week	Distance query; list/map; external navigation deep link; safety

Work Package	Complexity	Indicative Effort	Main Dependency / Completion Evidence
			wording.
Transparency/content module	Low-Medium	0.5-1 week	Published methodology/disclaimer/version content and admin editing.
Deployment	Medium	1-2 weeks	Render Django service; Render PostGIS database; GeoDjango native dependencies; static files; environment settings; staging/production smoke tests.
Android release	Medium	0.5-1 week	Signed Android App Bundle (AAB); Google Play internal/closed testing; production configuration.
Validation and refinement	High	Ongoing / final 2-4 weeks	Expert rule validation, GIS checks, DSS content review, usability/safety testing, defect fixes, documentation.

14. System-Wide Behavior and Acceptance Guide

14.1 Critical Consistency Rules

- The same published zone + rainfall scenario + rule-set version must produce the same susceptibility class on the dynamic map, Results Summary, Check My Area, explanation screen, and DSS context.
- Changing rainfall intensity or duration invalidates any old scenario-dependent result until it is re-evaluated.
- Changing the selected location invalidates a location-specific result until the new area is resolved/evaluated.
- Changing/publishing geographic data must not silently combine old polygons with incompatible new rule outputs.
- Only Approved/Published records may reach resident-facing API responses. Draft, Under Review, Rejected, and Superseded records remain non-public.
- Administrator geometry edits do not appear to users until the map/dataset version is published.
- The administrator cannot manually paint a public zone red/yellow/green to define susceptibility. Susceptibility color is derived from the Expert System result; admin edits the underlying approved knowledge/rules and geographic data.
- The DSS consumes the susceptibility result but never modifies it.
- DSS Preparedness Help returns only approved support topics/content. Unsupported questions receive a limitation/fallback instead of fabricated advice.
- Nearest evacuation center is a geographic distance ranking; it is not a declaration that the route is safe or the facility is open unless the verified center record explicitly states a current status supplied by the responsible authority.

14.2 Important Failure and Edge Cases

Failure / Edge Case	Required Behavior
Invalid email / duplicate email	Reject registration with a clear field-level message.
Incorrect login	Return a neutral Incorrect email or password message; never expose password details.
Expired mobile token	Attempt approved refresh; if refresh fails, return to login and preserve

Failure / Edge Case	Required Behavior
	only non-sensitive public state.
No rainfall scenario	Do not run assessment; direct user to select intensity and duration.
No Expert System rule matches	Return Insufficient Data with explanation and official-source reminder.
Equally applicable rules conflict	Return Uncertain; log the conflict for administrator/expert review; do not silently choose a class.
Pin outside study area	Return Outside Supported Area; do not classify.
Pin on ambiguous boundary	Ask user to adjust the pin or select from nearby supported areas.
Invalid polygon geometry	Block publication and identify the geometry that requires correction.
Missing source metadata	Block publication when source/provenance is mandatory for that rule/data type.
DSS branch has no transition	Block DSS version publication or show general approved fallback in a draft/test environment; never let a broken public tree guess a next step.
DSS loop / unreachable node	Flag during admin validation and block publish when it makes the public flow unsafe/incomplete.
Evacuation center unverified/outdated	Do not show as a normal active verified center; administrator must review/update/deactivate according to the defined policy.
Django/API unavailable	Show Service unavailable / Internet required. If a later approved cache exists, label it with date/version.
Basemap tiles unavailable	Keep FloodSense overlays/legend where feasible and show basemap unavailable; do not convert this into a susceptibility failure if the assessment itself succeeded.
New version published mid-session	Mark relevant assessment/DSS state stale and ask the user to refresh/re-evaluate before continuing.

14.3 Final System Boundaries

- FloodSense provides scenario-based flood susceptibility assessment and pre-event preparedness support for the approved Bacoar City study area.
- FloodSense is not a live flood sensor, real-time flood forecast, official warning service, current road-passability service, or independent evacuation authority.
- The Expert System uses expert-provided and expert-validated knowledge formalized into rules. It does not learn from resident behavior and does not automatically change its own rules.
- The dynamic map visualizes the Expert System output and verified map resources. It does not independently determine susceptibility.
- The DSS supports preparedness decisions through approved decision-tree logic and help topics. It does not replace emergency responders or official instructions.
- Geographic precision cannot exceed the resolution/accuracy of the approved source data.
- Conversational Artificial Intelligence is deferred and not required for the Minimum Viable Product (MVP).
- Administrative map/rule/DSS/content changes are controlled through draft, validation/review, publication, versioning, and audit history.

15. Implementation Checklist by Component

Component	Minimum Implementation Items
Flutter Android	Onboarding; register/login/reset; secure token handling; dashboard; rainfall input; http API client; flutter_map; polygon styling; map movement/zoom; zone popups; user pin; Check My Area; result explanation; Results Summary; DSS flow; DSS help topics; evacuation-center map/list; external navigation; profile/logout; error/stale states; signed release configuration.
Django Core	Custom email user model; admin/staff roles; Django sessions; password validators/reset; server configuration; templates/static files; system settings; error handling; audit utilities.
Django REST Framework (DRF)	Registration/login/token endpoints; profile; public config; GeoJSON zones; assessment; location resolution; map classification set; DSS start/answer/help; evacuation centers; version endpoint; serializer validation; permission classes; throttling.
GeoDjango	Polygon/MultiPolygon and Point fields; Spatial Reference System Identifier (SRID); geometry serialization; point-in-polygon; distance queries; spatial-index-aware queries; geometry validation support.
PostgreSQL/PostGIS	User/profile relations; barangays/zones; map versions; rainfall versions; zone profiles; expert rules/conditions/reviews; DSS nodes/options/transitions/help topics/actions; evacuation centers; source metadata; content versions; audit logs; constraints; migrations; indexes.
Admin Web	Django Admin for standard records; custom dashboard; Leaflet/Leaflet-Geoman map editor; GeoJSON import validation; rule manager/test sandbox; DSS editor/preview; rainfall manager; evacuation-center manager; source/provenance registry; content management; user support actions; version/publish/rollback; audit viewer.
Expert System	Knowledge-base data model; fact builder; forward chaining; match collector; conflict resolver; Low/Moderate/High/Very High; Uncertain/Insufficient Data; explanation facility; deterministic unit tests; batch validation cases.
Decision Support System (DSS)	Versioned tree; start-node logic; question/options; transitions; action retrieval; class-aware paths; supported help topics; safe fallback; path-validation tests; no susceptibility override.
Mapping	Basemap configuration; mobile layer style mapping; admin draw/edit/drag/delete; evacuation-center markers; geometry publication; map versioning; accessibility legend; no manual admin classification painting.
Deployment	Render PostgreSQL/PostGIS; Render Django Web Service; GeoDjango native libraries; environment variables; Unicorn; WhiteNoise/static files; HTTPS; production settings; staging; backups available under selected database plan; Android App Bundle; Google Play internal/closed testing.
Research and Validation	Formal data requests/permissions; final geographic resolution; expert knowledge elicitation; approved rule creation; validation cases; DSS preparedness-content review; evacuation-center verification; usability testing; safety/wording review; version/evidence archive.

16. Basis of This Revised Specification

This revised document uses the uploaded FloodSense materials as its project basis and then applies the updated design decisions supplied by the research team in this conversation. The earlier Revised Feature Specification remains the formatting/feature-flow guide. The feasibility study contributes the original project rationale, intended institutional knowledge sources, rainfall scenario concept, Check My Area, DSS, and

evacuation-center direction. The planning Software Requirements Specification contributes selected technical safeguards such as rule-based forward chaining, PostgreSQL/PostGIS, uncertainty/insufficient-data handling, provenance, validation, and version control. The present document is still a Feature Specification and development guide, not a Software Requirements Specification.

16.1 Project Source Files

- FloodSense_Feasibility_Study.pdf - feasibility rationale, flood-information problem, institutional data-source framing, PAGASA-aligned rainfall inputs, Expert System/DSS direction, Check My Area, evacuation resources, risks and mitigations.
- FloodSense_Revised_Feature_Specifications.pdf - primary feature-specification structure and the original end-to-end mobile flow.
- FloodSense_Revised_Feature_Specifications_SRS.docx - used selectively for the rule-based Expert System structure, forward chaining, PostgreSQL/PostGIS spatial design, uncertainty/insufficient-data states, provenance, validation, and change-control concepts.
- Updated research-team decisions - web administrator maintains the system; residents use Android; map is fully interactive and administratively editable; Expert System knowledge is supplied/validated by experts; DSS provides controlled decision support and supported preparedness answers; email + password is the mobile login; simplified Django-based technology stack is preferred when an extra framework is not required.

16.2 Current Official Technology Reference Notes

The technology recommendations were cross-checked against current official documentation while preparing this revision. The references below support the capabilities relied on in the architecture. Service pricing, quotas, and deployment details may change and must be rechecked immediately before public deployment.

Technology / Capability	Official Reference
Flutter Android build and release	https://docs.flutter.dev/deployment/android
Django GeoDjango with PostGIS	https://docs.djangoproject.com/en/5.2/ref/contrib/gis/install/postgis/
Django administration site	https://docs.djangoproject.com/en/5.2/ref/contrib/admin/
Django REST Framework authentication	https://www.django-rest-framework.org/api-guide/authentication/
Django REST Framework permissions	https://www.django-rest-framework.org/api-guide/permissions/
PostGIS point containment	https://postgis.net/docs/ST_Contains.html
PostGIS distance-within queries	https://postgis.net/docs/ST_DWithin.html
Leaflet-Geoman geometry editing	https://geoman.io/docs/leaflet
Render Django deployment	https://render.com/docs/deploy-django
Render PostgreSQL / PostGIS extension support	https://render.com/docs/postgresql

17. Final Recommended Architecture Summary

FloodSense should be implemented as one connected client-server system. Residents use a Flutter/Dart Android application. The mobile map is rendered with flutter_map. Administrators use a Django web administration system; conventional records can use Django Admin, while the editable GIS workspace uses custom Django views with Leaflet and Leaflet-Geoman. The same Django project exposes the mobile Application Programming Interface through Django REST Framework and performs spatial operations through GeoDjango. PostgreSQL with PostGIS stores the authoritative relational records, susceptibility

polygons, barangays, evacuation-center points, Expert System knowledge, DSS structure, versions, and audit history.

The Expert System is a deterministic Rule-Based Expert System using decision tables and forward chaining. Qualified/authorized experts and institutional sources provide or validate the domain knowledge; administrators encode and maintain that approved knowledge as versioned structured rules; the inference engine applies the rules to the selected rainfall scenario and geographic facts; the explanation facility states why the result was produced. The susceptibility result is then used by the dynamic map for color styling and by the Decision Support System as read-only decision context.

The Decision Support System is a separate deterministic preparedness layer. It uses the current Expert System result and the user's situational answers to follow approved branches and return appropriate pre-event preparedness actions. A DSS Preparedness Help feature allows users to choose supported questions/topics and retrieve approved answers. A conversational Artificial Intelligence assistant is not required for the MVP and should be considered only after the deterministic system is complete and validated.

The map is fully dynamic for the required scope: residents can pan, zoom, tap areas, position a pin, inspect susceptibility colors, and view evacuation centers; administrators can add/move point locations, draw/edit polygons, correct vertices, import GeoJSON, manage map layers, validate geometry, and publish new map versions. The administrator does not manually determine public susceptibility colors; the Expert System determines the class and the map translates the class into the published legend.

For deployment, use Render to host the Django Web Service and managed PostgreSQL/PostGIS database. Docker is recommended for the Django server package if needed to reliably include GeoDjango native dependencies such as GEOS, PROJ, and GDAL. The Android application is released as a signed Android App Bundle through Google Play. This architecture is intentionally simpler than a Flutter + Next.js + FastAPI + Firebase/Supabase combination while still supporting the complete FloodSense feature set described in this document.

Implementation Priority

Recommended order: 1) Django/PostGIS data model and authentication; 2) Django admin foundation and geographic map editor; 3) expert knowledge/rule schema and inference engine; 4) Flutter authentication/rainfall flow and dynamic map; 5) Results Summary and Check My Area; 6) DSS engine, help topics, and admin DSS editor; 7) evacuation-center management and nearest-center user flow; 8) versioning/audit/transparency; 9) staging validation; 10) Render deployment and Google Play testing; 11) optional offline caching or future conversational AI only after the core is stable.

18. Minimum Viable Product (MVP) vs. Deferred Enhancements

Category	Included in MVP / Core	Deferred Unless Later Required
User App	Onboarding; email/password account; dashboard; rainfall scenario; dynamic map; Check My Area; Expert System result/explanation; Results Summary; DSS; preparedness help; evacuation resources; account/logout.	Push notifications; background GPS; social login; crowdsourced reports; live traffic; live road closures.
Expert System	Expert-approved knowledge base; structured rules; decision tables; forward chaining; explanation; versioning; safe uncertainty states.	Machine Learning; automatic self-learning; external general-purpose AI classification.
DSS	Deterministic decision tree; user situational answers; approved actions; supported help topics/keyword search.	Open-domain chatbot; autonomous evacuation command; medical/real-time routing decisions.

Category	Included in MVP / Core	Deferred Unless Later Required
Map	Movable/zoomable map; dynamic susceptibility polygons; admin polygon/point editing; evacuation centers; manual pin; GeoJSON/PostGIS.	3D map; Augmented Reality; live radar; live sensor feed unless new scope/data are approved.
Admin	Authentication/roles; dashboard; map editor; rule manager/tester; DSS manager; rainfall; evacuation centers; sources; content; versioning; audit.	Complex organization workflow engine; separate React/Next.js portal; microservices.
Infrastructure	Django/DRF/GeoDjango; PostgreSQL/PostGIS; Leaflet/Leaflet-Geoman; Render; Google Play.	Redis/Celery; Kubernetes; separate Firebase/Supabase; separate object storage unless persistent attachments become required.